

Large Language Models in Education: A Literature Review

Bibliometric overview and synthesis of the literature since 2021

Data: 195 records, 2021–2026 (OpenAlex: large language models education)
Prepared with VOSStudio on 27 September 2026. Text written by an AI model from the papers listed; check it before use.

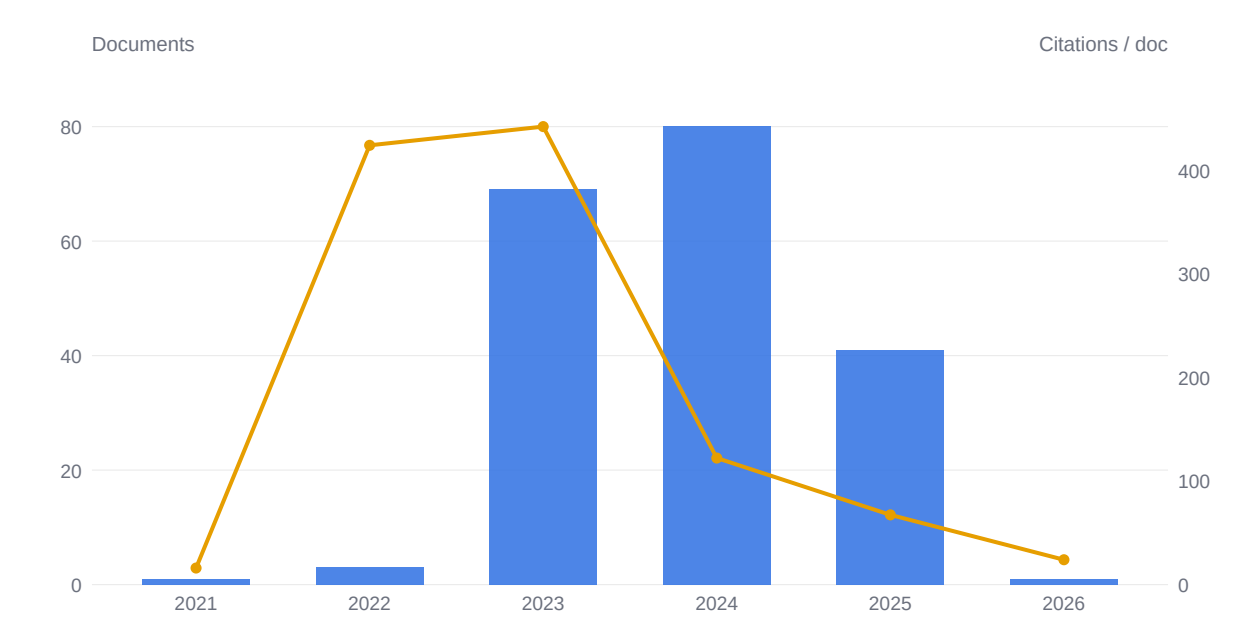


Figure 1. Annual scientific production. Number of documents per publication year

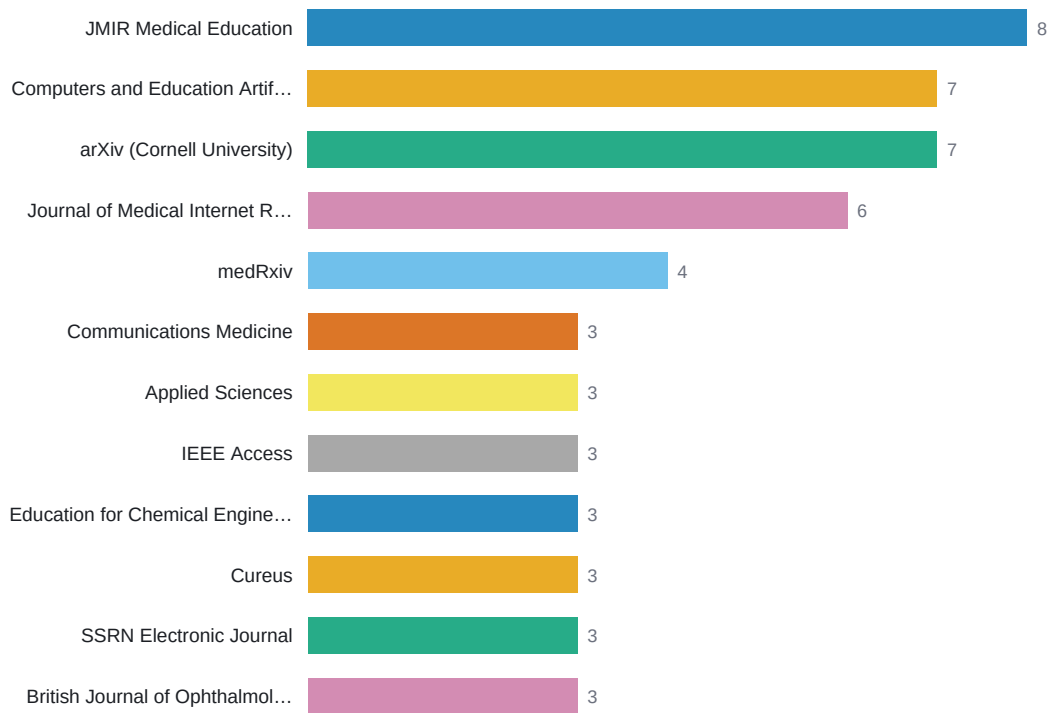


Figure 2. Most productive sources. Number of documents

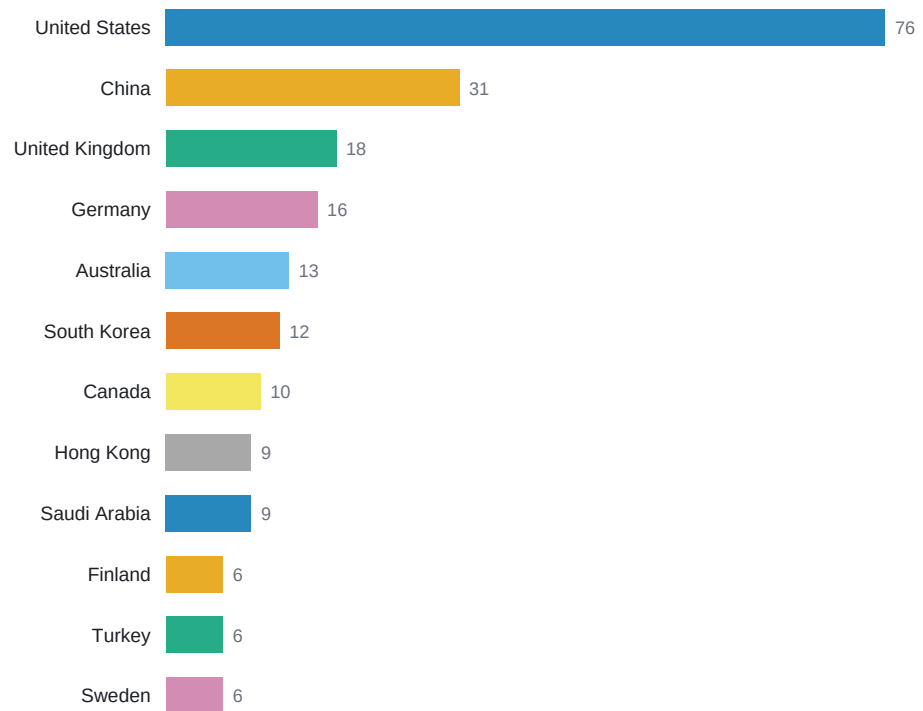


Figure 3. Most productive countries. Number of documents

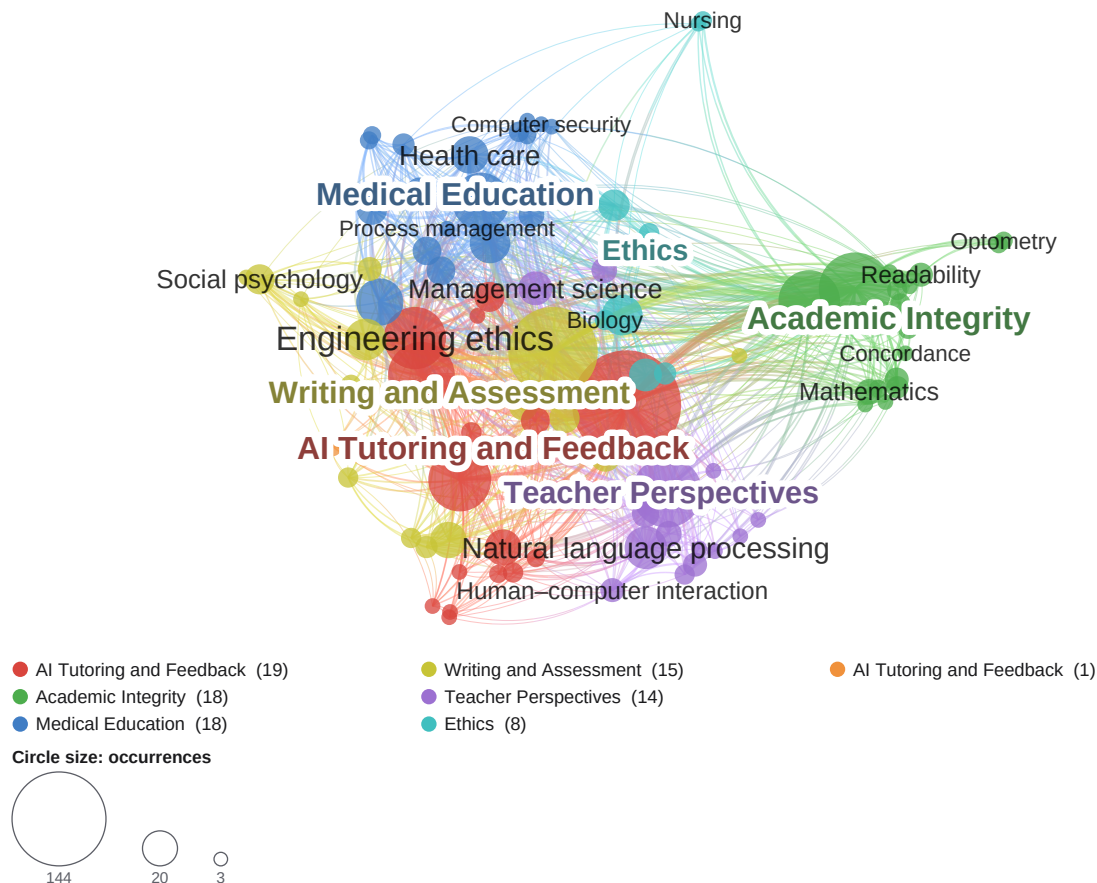


Figure 4. Network map: co-occurrence of all keywords. Keyword co-occurrence map; colours show the named clusters

Introduction

Large language models (LLMs) entered education quickly after 2022. This review summarises the literature indexed in OpenAlex since 2021 [1, 2]. Figure 1 shows the growth of the field, Figure 2 the main publication venues and Figure 3 the contributing countries.

The analysis combines publication statistics, a keyword co-occurrence map (Figure 4) and a reading of the most cited and most recent papers.

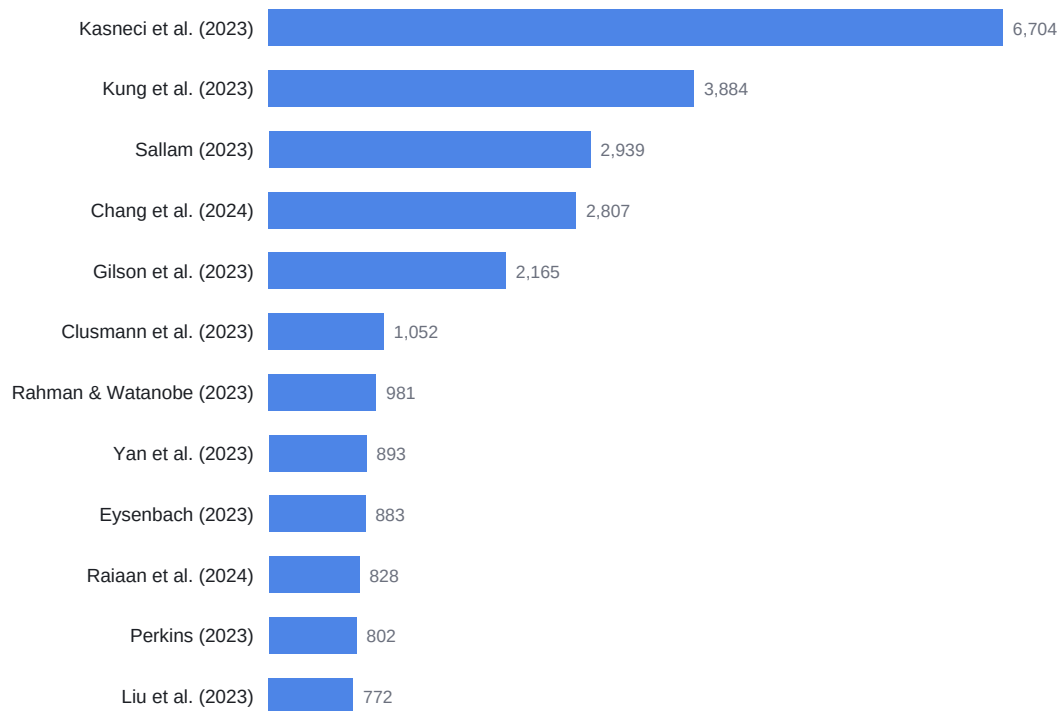


Figure 5. Most cited documents. Citations reported by the database

Research themes

AI tutoring and feedback

Several studies test LLMs as tutors and feedback providers [3, 4]. Results are promising for formative feedback.

Academic integrity

Concerns about plagiarism and assessment design are a second, highly cited strand [5].

- Detection tools remain unreliable [6].
- Assessment redesign is the preferred response [7–9].

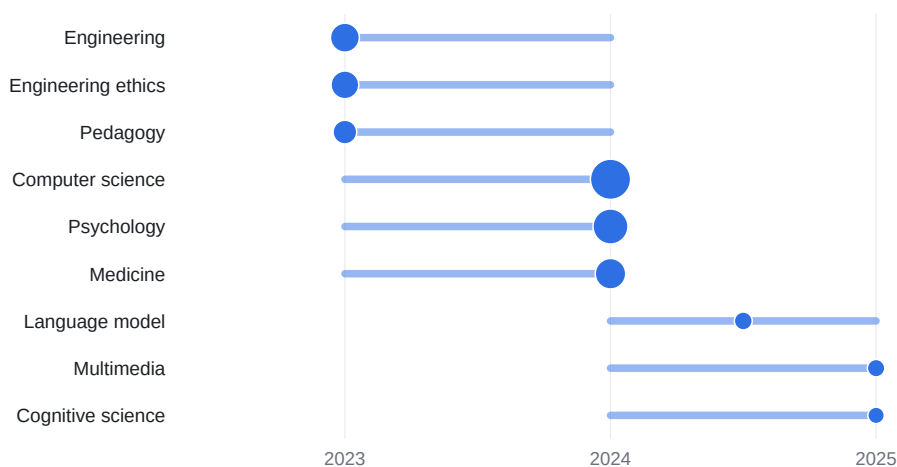


Figure 6. Trend topics (keywords). For each term, the line spans the first to third quartile of its publication years and the dot marks the median year; dot size shows frequency

Trends and emerging topics

Recent work moves from opinion pieces to empirical studies [10]. Figure 6 shows the recent terms.

Research gaps and future directions

1. Few longitudinal studies measure learning outcomes [11].
2. Evidence from low- and middle-income countries is scarce.
3. Teacher training is rarely evaluated.

Conclusion

The field grew fast and is consolidating around tutoring, integrity and assessment [1, 3].

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